

CS2601 Linear and Convex Optimization

Homework 1

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1. $\because \mathbf{A}\mathbf{v} = \begin{bmatrix} -6 \\ 0 \\ 2 \end{bmatrix} = 2\mathbf{v}$

$\therefore \mathbf{v}$ is the eigenvector of $\mathbf{A}$, and the associated eigenvalue is 2

$\because \mathbf{A}\mathbf{u} = \begin{bmatrix} 0 \\ 6 \\ 4 \end{bmatrix}$

$\therefore \mathbf{u}$ is not the eigenvector of $\mathbf{A}$

2. (a). $\nabla f(\mathbf{x}) = [f'(\mathbf{x})]^T = \mathbf{x}^T$

$\because [\nabla^2 f(\mathbf{x})]_{ij} = \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j}$

$\therefore$ If and only if $i = j$, $[\nabla^2 f(\mathbf{x})]_{ij} = 1$, otherwise $[\nabla^2 f(\mathbf{x})]_{ij} = 0$

$\therefore [\nabla^2 f(\mathbf{x})] = \mathbf{E}$

(b). $\nabla f(\mathbf{w}) = [f'(\mathbf{w})]^T = [\mathbf{X}(\mathbf{X}\mathbf{w} - \mathbf{y}) + \lambda\mathbf{w}]^T = \mathbf{w}^T(\mathbf{X}^2)^T - \mathbf{y}^T\mathbf{X}^T + \lambda\mathbf{w}^T$

$[\nabla^2 f(\mathbf{x})] = f''(\mathbf{w}) = \mathbf{X}^2 + \lambda$

3. $\because \mathbf{A}$ is positive semidefinite.

$\therefore$

$$\begin{cases} \det \mathbf{A}_{\{2,3\}} = 2 - \alpha^2 \geq 0 \\ D_3(\mathbf{A}) = -3\alpha^2 - 4\alpha \geq 0 \end{cases}$$

$\therefore$ When $\alpha \in [-\sqrt{2}, 0]$, $\mathbf{A}$ is positive semidefinite.

4. (a). Let $\nabla f(\mathbf{x}) = [4x_1 + 2x_2 + 2, 5x_2 + 2x_1 - 2x_3 - 3, 6x_3 - 2x_2 + 2] = \mathbf{0}$

Then $x_1 = -1, x_2 = 1, x_3 = 0, \mathbf{x}^* = [-1, 1, 0]^T$

The stationary point is $\mathbf{x}^* = [-1, 1, 0]^T$

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} 4 & 2 & 0 \\ 2 & 5 & -2 \\ 0 & -2 & 6 \end{bmatrix}$$

$\because D_1(\nabla^2 f(\mathbf{x})) = \det(4) = 4 > 0, D_2(\nabla^2 f(\mathbf{x})) = \det \begin{bmatrix} 4 & 2 \\ 2 & 5 \end{bmatrix} = 16 > 0, D_3(\nabla^2 f(\mathbf{x})) =$

$\det \nabla^2 f(\mathbf{x}) = 80 > 0$

$\therefore \nabla^2 f(\mathbf{x})$ is positive definite. Then $\mathbf{x}^*$ is local minima

(b). Let $\nabla f(\mathbf{x}) = [x_1 + 2x_2, 2x_2 + 2x_1 - x_3 + 1, -3x_3 - x_2 - 3] = \mathbf{0}$

Then $x_1 = -\frac{12}{5}, x_2 = \frac{6}{5}, x_3 = -\frac{7}{5}, \mathbf{x}^* = [-\frac{12}{5}, \frac{6}{5}, -\frac{7}{5}]^T$

The stationary point is $\mathbf{x}^* = [-\frac{12}{5}, \frac{6}{5}, -\frac{7}{5}]^T$

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 2 & -1 \\ 0 & -1 & -3 \end{bmatrix}$$

$$\because D_1(\nabla^2 f(\mathbf{x})) = \det(1) = 1 > 0, D_2(\nabla^2 f(\mathbf{x})) = \det \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix} = -2 < 0$$

$\therefore \nabla^2 f(\mathbf{x})$ is neither positive definite nor negative definite. Then $\mathbf{x}^*$ is neither local minima nor local maxima

5. (a) $\because f(\mathbf{w}) = \sum_{i=1}^m \log(1 + e^{-y_i \mathbf{x}_i^T \mathbf{w}}) > 0$

Then let $\mathbf{w} = N\mathbf{w}_0, N \in N^+, \forall i = 1, 2, \dots, m, y_i \mathbf{x}_i^T \mathbf{w} > 0$

The let $N \rightarrow +\infty, \log(1 + e^{-y_i \mathbf{x}_i^T \mathbf{w}}) \rightarrow 0, f(\mathbf{w}) \rightarrow 0.$

$\therefore f$ doesn't have a global minimum.

(b). (i) Suppose $h(\mathbf{w}) = -y_a \mathbf{x}_a^T \mathbf{w}$

Then $f(\mathbf{w}) = \sum_{i=1}^m \log(1 + e^{-y_i \mathbf{x}_i^T \mathbf{w}}) \geq \log(1 + e^{-y_a \mathbf{x}_a^T \mathbf{w}}) \geq \log(e^{-y_a \mathbf{x}_a^T \mathbf{w}}) = -y_a \mathbf{x}_a^T \mathbf{w} = h(\mathbf{w})$

$\therefore f(\mathbf{w}) \geq h(\mathbf{w})$

(ii) $\because \forall \mathbf{w}, \exists i_0 = 1, 2, \dots, m$, such that $y_{i_0} \mathbf{x}_{i_0}^T \mathbf{w} < 0$

$\therefore h(\mathbf{w}) > 0$

$\therefore$ If $h(\mathbf{w})$ has a minima, then $C = h(\mathbf{w}_0) > 0$

$\because \|\mathbf{w}\| = 1,$

$\therefore |h(\mathbf{w})| = |y_i \mathbf{x}_i^T \mathbf{w}| = |\mathbf{x}_i^T \mathbf{w}| < \sum_{i=1}^m \|\mathbf{x}_i\|$

$\therefore h(\mathbf{w})$ is bounded.

$\therefore h(\mathbf{w})$ is continuous.

$\therefore h(\mathbf{w})$ has a global minimum $\mathbf{w}_0$ on S and $C = h(\mathbf{w}_0) > 0$

(iii) $\because h(\mathbf{w}) = \max_{1 \leq i \leq m} -y_i \mathbf{x}_i^T \mathbf{w}$

$\therefore h(\mathbf{w}) = \|\mathbf{w}\|(-y_a \mathbf{x}_a^T \frac{\mathbf{w}}{\|\mathbf{w}\|})$

$\forall i = 1, 2, \dots, m, h(\mathbf{w}) \geq \|\mathbf{w}\|(-y_i \mathbf{x}_i^T \frac{\mathbf{w}}{\|\mathbf{w}\|}) \geq h(\mathbf{w}_0)\|\mathbf{w}\| = C\|\mathbf{w}\|$

(iv) $\because f(\mathbf{w}) \geq h(\mathbf{w}) \geq C\|\mathbf{w}\|$

$\therefore$ When $\|\mathbf{w}\| \rightarrow +\infty, C\|\mathbf{w}\| \rightarrow +\infty, f(\mathbf{w}) \rightarrow +\infty$

$\therefore f(\mathbf{w})$ is continuous

$\therefore f(\mathbf{w})$ has a minima

(c). $\nabla f(\mathbf{w}) = (f'(\mathbf{w}))^T = (\sum_{i=1}^m \frac{-e^{-y_i \mathbf{x}_i^T \mathbf{w}} y_i \mathbf{x}_i^T}{1 + e^{-y_i \mathbf{x}_i^T \mathbf{w}}})^T = \sum_{i=1}^m -\mathbf{x}_i \frac{y_i}{1 + e^{y_i \mathbf{x}_i^T \mathbf{w}}}$